Slick Trace investigation report — case_0006

Generated 2026-09-25T18:33:57.400942+00:00 · reproducible investigation report

What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

Finding

No vessel in this scene can account for the observed slick. No candidate release reproduced the observation above the absolute fit gate. The system returns no name rather than the least-bad one; a source vessel may have left the scene before acquisition.

Scene

Field	Value
Incident	case_0006
Acquired (UTC)	2026-09-10T04:30:00+00:00
Sensor	synthetic SAR-like (VV proxy, single channel)
Data status	synthetic
Pixel spacing	433.59 m
Label source	generator ground truth (synthetic) — not a human or independently confirmed label

Evidence layers

Layer	Provenance
AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.

Traffic filtering

State	Vessels
unassessed	2
candidate	3

Unassessed is reported separately from excluded: missing evidence is not exoneration, and collapsing the two would silently convert one into the other.

Rank fragility

Leader retained in 0.5125 of 80 tested perturbations. A tested perturbation changed the leader; the smallest is recorded in the JSON report.

Absence of a flip is evidence of stability across the tested set, not a proof of stability: single-axis and two-axis perturbations only.

File integrity

File	SHA-256 (first 16)
attribution_result.json	8ef0b0ed199a1e5a
coverage_result.json	d1ea8c538c42cbdb
forcing.npz	07595f69999ce4c5
forecast_result.json	6f3ce20d79caf404
fragility_result.json	8a896e89c0f1a439
historical_mapping.json	92a02eb5ab4b1dc6
manifest.json	2d6f27e9ab4d9c73
mask_groundtruth.png	e424c14c491b48c9
naive_baseline_result.json	16bd5fc0659fb747
next_observation_result.json	378c277f832846b5
scene.png	36d4a1e0d7b61922

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

Uncertainty

• Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

• Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

• Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

• Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

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